

Problem 15.15

An annuity provides a payment of n at the end of each year for n years. The annual effective interest rate is $\frac{1}{n}$. What is the present value of the annuity?

Solution

The annual effective interest rate is

$$i = \frac{1}{n}.$$

Since the annuity pays n at the end of each year for n years, its present value is

$$PV = na_{\overline{n}|i}.$$

Using the annuity-immediate formula

$$a_{\overline{n}|i} = \frac{1 - v^n}{i},$$

we get

$$PV = n \left(\frac{1 - \left(1 + \frac{1}{n}\right)^{-n}}{\frac{1}{n}} \right).$$

Therefore,

$$\begin{aligned} PV &= n^2 \left(1 - \frac{1}{\left(1 + \frac{1}{n}\right)^n} \right) \\ &= n^2 \left(1 - \left(\frac{n}{n+1} \right)^n \right). \end{aligned}$$

Hence, the present value of the annuity is

$$\boxed{n^2 \left(1 - \left(\frac{n}{n+1} \right)^n \right)}.$$